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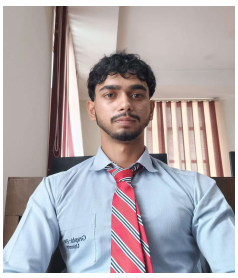
PROJECT AND TEAM INFORMATION

Project Title

CrisisConnect: Linking help saving lives

Student / Team Information

Team Name: Team #	CrisisCoders FS-VI-T090
Team member 1 (Team Lead)	Dhoundiyal, Aryan – 23011562 23011562@geu.ac.in
Team member 2	Rawat,Vaibhav *– 23021661* *23021661@geu.ac.in* 

Team member 3	Negi, Harshit – 23021720 23021720@geu.ac.in

PROJECT PROGRESS DESCRIPTION

Project Abstract

Natural disasters create major challenges for both communication and coordination in the real living world. Disaster affected people(victims) require essential resources like food, water, shelter or rescue support on an urgent basis. But mostly at disaster affected areas, the information is generally too scattered and unorganized. Victims often also lose their lives due to the delayed information.
So we need a system where all these aspects are kept in mind and are organized. And this is where our web-based platform comes in picture.
CrisisConnect is a web-based system which enables disaster management authorities and volunteer teams to coordinate and carry out relief operations. The platform uses structured request management and volunteer assignment and centralized monitoring to enhance response efficiency during emergency situations.

Updated Project Approach and Architecture

- We will follow a modular and phase-wise approach to complete this project.
- Firstly , we will define the requirements for this project like who will be the primary users for our platform , what will be the modules to be developed , and what are the core features in our project.
- Before the coding/implementation phase , we will design the system architecture and database schema for our platform.
- Now, the module wise development will be there.
- This development will be further be divided into phases.
- Integration will be done simultaneously with the development of other phases as well.
- Once the coding part is complete, testing and validation phase will begin.
- Manual UI testing, unit testing , API testing will be the part of this phase.
- Version Control will be used for the collaboration purpose.
- Future works may include advanced analytics, resource inventory tracking, real-time updates and SMS integration.

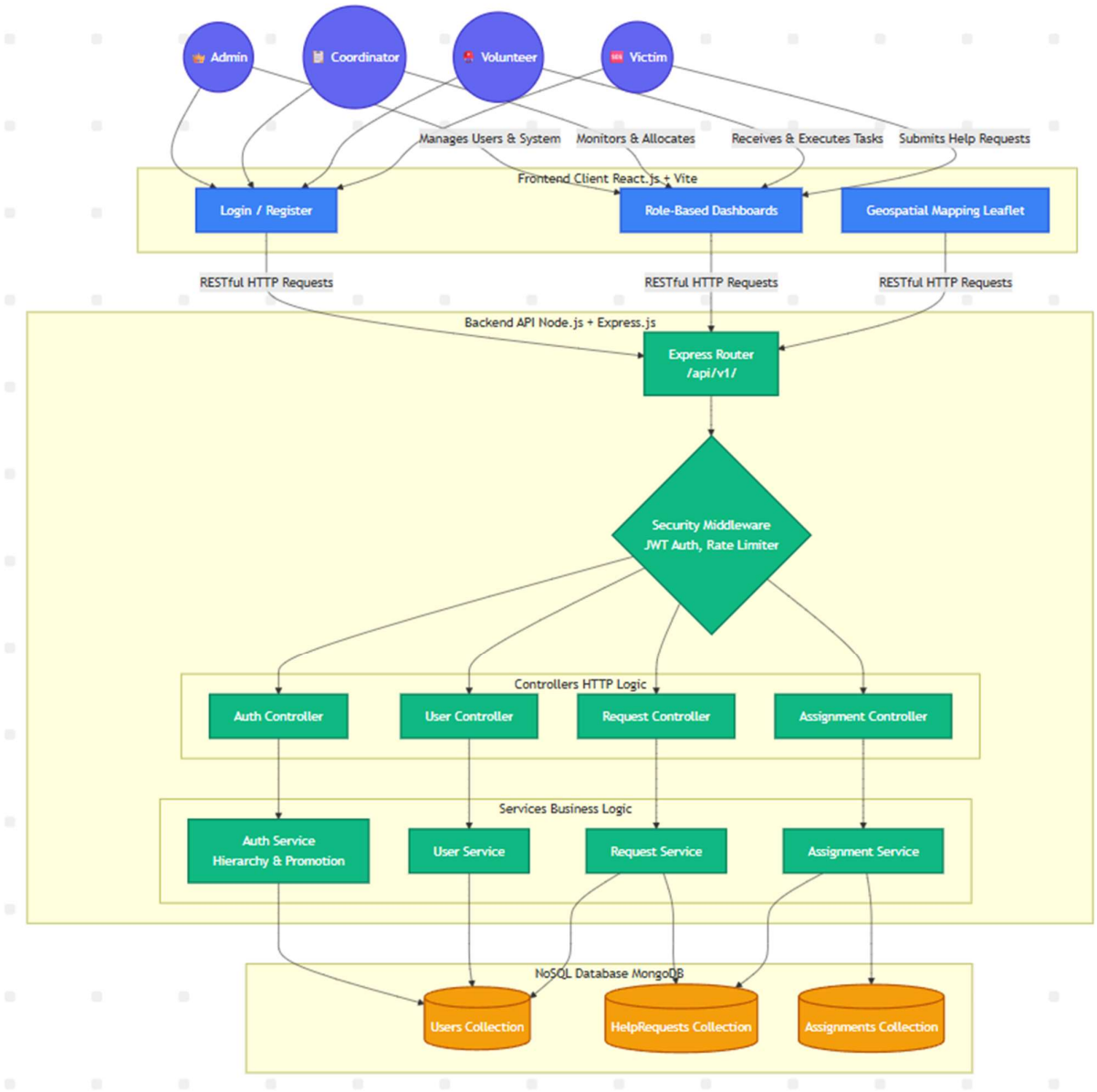


Fig 1.1 System architecture for CrisisConnect

Tasks Completed

Task Completed	Team Member
Frontend UI Components Development	Harshit Negi
Backend API Development	Aryan Dhoundiyal
MongoDB Database Schema Design	Vaibhav Rawat
Basic Map Integration	Vaibhav Rawat
Authentication System (Backend)	Aryan Dhoundiyal
Offline Sync Module	Vaibhav Rawat
UI Polishing and Refinement	Harshit Negi
Comprehensive Testing Suite	All team members
Project Repository Setup	All team members

Challenges/Roadblocks

The main challenge during the implementation of this project is to integrate the frontend with the backend while maintaining the correctness of authentication system done with the help of JWT(Java Web Tokens).
The primary challenge was then shifted to the backend implementation. The major challenge was building the RESTful APIs in Node.js that would connect the user interface and the database. A significant task within this was implementing a secure user authentication system, which involved password hashing and protecting routes to prevent unauthorized access.
Maintaining a responsive and user-friendly system required multiple iterations and coordination between the members.
Moreover , team members working over different independent components required a proper coordination between them and required frequent communication so that the end interface would align as per the final project.

Project Outcome/Deliverables

- A fully developed web application on MERN stack technology.
- Implementation of role - based authentication for different roles.
- A fully functional web platform for handling disaster-related help requests.
- A dashboard that displays the details and status of the requests.
- The source code for the web application.
- Practical experience gained in building web applications.
- The unit tests for the major components.
- The final project report and demonstration.

Progress Overview

- The user interface and the dashboard for each roles i.e. Admin , Volunteer , Victims has been fully implemented.
- Backend APIs have been fully completed and implemented and are being tested.
- Database Connectivity (MongoDB) has been implemented and is successfully connected with the backend.
- Integration between frontend and backend is done and testing is almost completed.
- Secure JWT based authentication has been completed now with proper password hashing and preventing unauthorized user access.
- Map integrations in different dashboards are now implemented and functional.

Codebase Information

```
CrisisConnect/
├── Crisis_backend/        # Express.js Server
│   ├── config/            # Database & environment configurations
│   ├── controllers/       # Logic for handling API requests
│   ├── middleware/        # Auth guards & error handlers
│   ├── models/            # Database schemas (User, Request, Assignment)
│   ├── routes/            # API endpoint definitions
│   ├── services/          # Reusable business logic & DB operations
│   ├── utils/             # Helper utilities (tokens, validators)
│   ├── server.js          # Entry point for the backend
```

```
│
├── Crisis_frontend/        # React (Vite) Application
│   ├── public/             # Static assets (images, icons)
│   ├── src/
│   │   ├── api/            # API integration (Axios instances)
│   │   ├── components/     # Common UI components (Buttons, Modals)
│   │   ├── context/        # Global state (AuthContext, ThemeContext)
│   │   ├── pages/          # Views (Dashboard, Home, Login)
│   │   ├── styles/         # Global CSS and themes
│   │   ├── App.jsx         # Main routing logic
│   │   └── main.jsx        # React mounting point
```

Github Repository : https://github.com/Aryan07-afk/CrisisConnect
Branch : main

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Unit Tests - Frontend Components	**Pass**	Basic component tests implemented
Unit Tests - Backend APIs	**Pass**	API endpoint tests ongoing
Integration Tests	**Pass**	Completed
Database Schema Validation	**Pass**	Schema verified and collections created
User Authentication Test	**Pass**	Backend tests passing
Map Integration Test	**Pass**	Basic map rendering functional
End-to-End Testing	**Pass**	Major tests passed

Deliverables Progress

- **Interactive Dashboard for each roles i.e. Admin , Volunteer , Victims :**
  Frontend UI : Completed
- **Backend APIs :** Completed
- **Schema Design and Database Integration :** Completed
- **Testing :** Completed
- **Map Integration :** Completed